RELATÓRIO TÉCNICO: INSTALAÇÃO E CONFIGURAÇÃO DO SERVIDOR SAMBA

Disciplina: Administração e Serviços de Rede

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1. Identificação de Rede do Servidor

Antes de iniciar a instalação, foi executado o comando para identificar as interfaces de rede e o endereço IP do servidor.

- **Comando utilizado:** ip a
- **IP do Servidor Identificado:** 172.30.1.2 na interface enp1s0.

```
Editor   Tab 1   Tab 2   +
root@ubuntu:~$ ^[[200~
: command not found
root@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc fq_codel state UP group default qlen 1000
    link/ether f6:9b:19:e9:d8:d2 brd ff:ff:ff:ff:ff:ff
    inet 172.30.1.2/24 brd 172.30.1.255 scope global dynamic noprefixroute enp1s0
       valid_lft 86311727sec preferred_lft 75522527sec
    inet6 fe80::9a63:22bd:cbef:14ec/64 scope link
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1454 qdisc noqueue state DOWN group default
    link/ether 76:92:24:03:ef:60 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
```

2. Instalação do Serviço Samba

Com o IP anotado, o repositório local de pacotes foi atualizado e o Samba foi instalado no sistema.

```
sudo apt update
sudo apt install samba -y
```

Para garantir que o servidor subiu corretamente, verificou-se o status do daemon smbd:

```
sudo systemctl status smbd
```

O serviço foi reportado em execução ativa (**active (running)**).

```
Editor   Tab 1   Tab 2   +
       valid_lft forever preferred_lft forever
root@ubuntu:~$ ^[[200~sudo apt update
sudo: command not found
root@ubuntu:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
17 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ubuntu:~$ sudo apt install samba -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  squashfs-tools
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  attr libavahi-client3 libavahi-common-data libavahi-common3 libboost-iostreams1.83.0
  libtalloc2 libtdb1 libtevent0t64 liburing2 libwbclient0 python3-dnspython python3-gp
  samba-common samba-common-bin samba-dsdb-modules samba-libs samba-vfs-modules tdb-to
Suggested packages:
  cups-common python3-trio python3-aioquic python3-h2 python3-httpx python3-httpcore p
The following NEW packages will be installed:
  attr libavahi-client3 libavahi-common-data libavahi-common3 libboost-iostreams1.83.0
  libtalloc2 libtdb1 libtevent0t64 liburing2 libwbclient0 python3-dnspython python3-gp
  samba-common samba-common-bin samba-dsdb-modules samba-libs samba-vfs-modules tdb-to
0 upgraded, 32 newly installed, 0 to remove and 17 not upgraded.
```

```
root@ubuntu:~$ sudo systemctl status smbd
● smbd.service - Samba SMB Daemon
     Loaded: loaded (/usr/lib/systemd/system/smbd.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-05-28 23:29:13 UTC; 16s ago
       Docs: man:smbd(8)
             man:samba(7)
             man:smb.conf(5)
   Main PID: 2738 (smbd)
     Status: "smbd: ready to serve connections..."
      Tasks: 3 (limit: 2237)
     Memory: 10.4M (peak: 10.6M)
        CPU: 75ms
     CGroup: /system.slice/smbd.service
             ├─2738 /usr/sbin/smbd --foreground --no-process-group
             ├─2741 "smbd: notifyd" .
             └─2742 "smbd: cleanupd "

May 28 23:29:13 ubuntu systemd[1]: Starting smbd.service - Samba SMB Daemon...
May 28 23:29:13 ubuntu (smbd)[2738]: smbd.service: Referenced but unset environment variable evaluates to an empty string: SMBDOPTIONS
May 28 23:29:13 ubuntu systemd[1]: Started smbd.service - Samba SMB Daemon.
```

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3. Criação de Diretórios e Permissões no Sistema

Foram preparados os diretórios locais para os compartilhamentos. No exemplo da pasta privada, aplicou-se a restrição de donos e permissões para que apenas o usuário autorizado tenha acesso total:

```
sudo chown ubuntu:ubuntu /srv/samba/privado
sudo chmod 700 /srv/samba/privado
```

```
root@ubuntu:~$ sudo chown ubuntu:ubuntu /srv/samba/privado
root@ubuntu:~$ sudo chmod 700 /srv/samba/privado
root@ubuntu:~$ sudo cp /etc/samba/smb.conf /etc/samba/smb.conf.bak
root@ubuntu:~$ sudo nano /etc/samba/smb.conf
root@ubuntu:~$
```

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4. Configuração dos Compartilhamentos (smb.conf)

Antes de editar o arquivo de configuração principal, foi gerada uma cópia de segurança (backup):

```
sudo cp /etc/samba/smb.conf /etc/samba/smb.conf.bak
```

O arquivo foi aberto com o editor nano e, ao final dele, foram mapeadas as regras para a **Pasta Pública** e a **Pasta Privada**:

```
[publico]
  comment = Pasta Publica - Sem Senha
  path = /srv/samba/publico
  browsable = yes
  guest ok = yes
  read only = no
  force user = nobody

[privado]
  comment = Pasta Privada
  path = /srv/samba/privado
  browsable = yes
  guest ok = no
  read only = no
  valid users = ubuntu
```

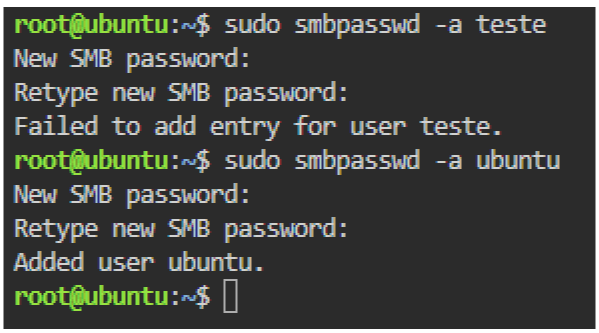
Editor Tab 1 Tab 2 +

```
 GNU nano 7.2                                                    /etc/samba/smb.conf *
;[profiles]
;   comment = Users profiles
;   path = /home/samba/profiles
;   guest ok = no
;   browseable = no
;   create mask = 0600
;   directory mask = 0700

[printers]
   comment = All Printers
   browseable = no
   path = /var/tmp
   printable = yes
   guest ok = no
   read only = yes
   create mask = 0700

# Windows clients look for this share name as a source of downloadable
# printer drivers
[print$]
   comment = Printer Drivers
   path = /var/lib/samba/printers
   browseable = yes
   read only = yes
   guest ok = no
# Uncomment to allow remote administration of Windows print drivers.
# You may need to replace 'lpadmin' with the name of the group your
# admin users are members of.
# Please note that you also need to set appropriate Unix permissions
# to the drivers directory for these users to have write rights in it
;   write list = root, @lpadmin

[publico]
   comment = Pasta Publica - Sem Senha
   path = /srv/samba/publico
   browsable = yes
   guest ok = yes
   read only = no
   force user = nobody

[privado]
   comment = Pasta Privada - Apenas seu_usuario
   path = /srv/samba/privado
   browsable = yes
   guest ok = no
   read only = no
   valid users = ubuntu
```

5. Cadastro de Usuários e Ajuste do Firewall

O Samba mantém um banco de dados de senhas próprio. O usuário ubuntu foi adicionado e sua respectiva senha de rede criada. Logo após, as portas do protocolo foram liberadas no firewall nativo do Linux (ufw).

```
sudo smbpasswd -a ubuntu
sudo ufw allow Samba
```



```
root@ubuntu:~$ sudo smbpasswd -a teste
New SMB password:
Retype new SMB password:
Failed to add entry for user teste.
root@ubuntu:~$ sudo smbpasswd -a ubuntu
New SMB password:
Retype new SMB password:
Added user ubuntu.
root@ubuntu:~$
```

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6. Testes de Conectividade e Acesso (Máquina Cliente)

Em uma estação de trabalho cliente, instalaram-se as ferramentas de teste de rede:

```
sudo apt install smbclient cifs-utils -y
```

6.1. Mapeamento de Recursos Disponíveis

Para listar o que o servidor (172.30.1.2) estava oferecendo de forma anônima (parâmetro -N):

```
smbclient -L //172.30.1.2 -N
```

O prompt listou com sucesso os volumes publico e privado mapeados.

Editor Tab 1 Tab 2 +

```
root@ubuntu:~$ sudo apt install smbclient cifs-utils -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  squashfs-tools
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  keyutils libsmbclient0
Suggested packages:
  winbind heimdal-clients
The following NEW packages will be installed:
  cifs-utils keyutils libsmbclient0 smbclient
0 upgraded, 4 newly installed, 0 to remove and 17 not upgraded.
Need to get 686 kB of archives.
After this operation, 2886 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 cifs-utils amd64
 2:7.0-2ubuntu0.2 [96.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 keyutils amd64 1.6.3-3bu
ild1 [56.8 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libsmbclient0 am
d64 2:4.19.5+dfsg-4ubuntu9.6 [62.4 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 smbclient amd64
2:4.19.5+dfsg-4ubuntu9.6 [470 kB]
```

```
root@ubuntu:~$ smbclient -L //172.30.1.2 -N

        Sharename       Type      Comment
        ---------       ----      -------
        print$          Disk      Printer Drivers
        publico         Disk      Pasta Publica - Sem Senha
        privado         Disk      Pasta Privada - Apenas seu_usuario
        IPC$            IPC       IPC Service (ubuntu server (Samba, Ubuntu))
SMB1 disabled -- no workgroup available
root@ubuntu:~$
```

6.2. Teste de Acesso à Pasta Pública

O acesso à pasta pública foi testado de forma direta e sem credenciais:

`smbclient //172.30.1.2/publico -N`

Conexão bem-sucedida, abrindo o prompt interativo smb: \>.

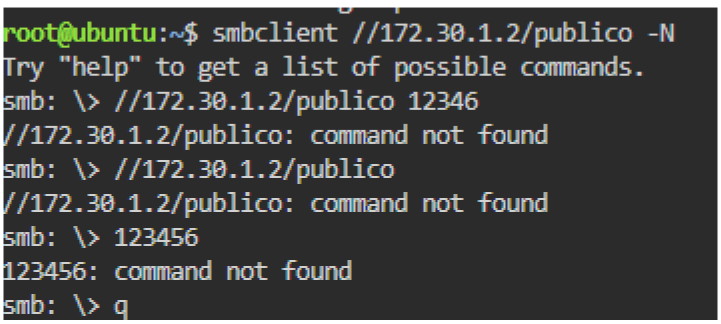
6.3. Teste de Acesso à Pasta Privada

O acesso à pasta restrita foi validado passando explicitamente o usuário autorizado (-U ubuntu):

```
smbclient //172.30.1.2/privado -U ubuntu
```

Após digitar a senha de rede definida, o cliente obteve acesso direto ao prompt interno do diretório privado (smb: \>), consolidando o sucesso da atividade.

Pasta Pública



Pasta privada